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ORIGINAL PAPER

Artificial Intelligence in Higher Education: Early Perspectives from Lebanese STEM Faculty

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Abstract

This study examines the integration of artificial intelligence (AI) chatbots within higher education, focusing on the early feedback from STEM faculty members in Lebanon, particularly in the context of their widespread use among students. As AI intersects with traditional scientific inquiry, its incorporation into educational frameworks introduces both opportunities and challenges. This research aimed to explore faculty perceptions of AI chatbot usage, assess the benefits and challenges of their adoption, and provide insights to inform AI integration strategies in higher education, particularly in developing countries like Lebanon. This study employs a mixed-methods approach, combining qualitative and quantitative data collection to evaluate STEM faculty members' perspectives on AI chatbots. Surveys were distributed via email to 30 STEM higher educators in Lebanon. The surveys included quantitative components (Likert-scale and multiple-choice) and qualitative open-ended questions to capture nuanced feedback. Faculty members demonstrated proactive engagement with AI tools, acknowledging their potential to enhance accessibility, improve efficiency, and support academic tasks. However, concerns about risks such as over-reliance, misinformation, and ethical dilemmas were raised. The findings underscore the need for AI-specific guidelines and greater AI literacy among educators to maximize benefits while mitigating challenges. AI chatbots present significant opportunities for advancing higher education but require careful integration to align with academic goals and preserve integrity. By emphasizing AI literacy and tailored guidelines, this study offers a roadmap for leveraging AI in Lebanese higher education and similar contexts in developing countries.

Keywords AI Integration in Education · STEM Faculty · Educational Technology Challenges · AI Chatbots in Higher Education · Lebanon

Introduction

The integration of artificial intelligence (AI) chatbots into higher education represents a pivotal shift, particularly for STEM educators, as they stand at the crossroads of rapid technological advancements and long-standing academic traditions (Clark et al., 2024; Mhlanga, 2023; Yik & Dood, 2024). Recent developments in generative AI, notably the emergence of new AI chatbots between late 2022 and early 2023 (Mahrishi et al., 2024), have sparked significant interest across public and academic sectors (Mollick & Mollick, 2024; Ramirez & Esparrell, 2023; Waltzer et al., 2023; Zawacki-Richter et al., 2019). These tools are redefining traditional tasks such as essay writing, lesson planning, and the customization of student assignments, thus catalyzing a wider discourse on AI's role within educational frameworks. While students, particularly those from Generation Z (born between 1997 and 2012), are quickly adopting and utilizing AI chatbots, faculty members' adoption rates vary, with some lagging behind their pupils. Research done by Lozano and Blanco Fontao at the University of Leon, illustrates a generally optimistic student perspective towards AI integration in education, particularly when the authenticity of data

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sources is ensured (Lozano & Blanco Fontao, 2023). Similarly, students in the Middle East have expressed positive attitudes toward AI tools like ChatGPT, drawn by their ease of use and perceived efficacy in educational settings (Abdaljaleel et al., 2024). However, for STEM faculty, the advent of AI presents a dual-edged sword of remarkable opportunities and significant challenges. AI has the potential to streamline assessments, enhance teaching methods, and diversify assessment strategies offering educators time-saving benefits and deeper insights into student performance (Nikolic et al., 2023). Despite these advantages, faculty in research-driven fields remain cautious about adopting AI, driven by concerns over data privacy, academic integrity, and the risk of reinforcing biases. Educators are also confronted with significant concerns about data reliability, ethical dilemmas, and the preservation of academic integrity (Su & Yang, 2023; Zhai & Nehm, 2023). This is underscored in the comprehensive review by Imran and Almusharraf (2023), which acknowledged the substantial utility of OpenAI's ChatGPT in academic composition across various disciplines.

Research Questions

Given the lack of research on Lebanese faculty's engagement with AI tools, this study aims to offer preliminary assessment of how AI chatbots are being integrated within Lebanon's STEM higher education sector, while focusing on the unique challenges and opportunities within a developing country context. The research questions formulated for this study seek to explore the extent of AI chatbots adoption, the perceived benefits, and the obstacles encountered by STEM educators in Lebanon as they integrate them into their academic and research practices:

1. How rapidly are faculty members in Lebanon's STEM fields adapting to and engaging with recent developments in AI technology, and what factors, such as age, major, and teaching experience, could influence this adaptation?
2. What specific benefits have Lebanese STEM faculty recognized from using AI technologies, and how do they assess the impact of these tools on their academic and research activities?

The findings can inform AI adoption strategies not only in Lebanon but also in other developing countries with similar educational and technological landscapes. Furthermore, this study adds to the international discourse on AI's role in education by exploring faculty perspectives in regions that are often underrepresented in AI research, thereby offering a more comprehensive understanding of the global challenges and opportunities associated with AI integration in academia.

Research Methodology

Research Design

This study employs a mixed-method research methodology to explore early feedback on the integration of AI chatbots in the work of higher education instructors and to gather their perspectives on these tools (Creswell & Creswell, 2018). The data collection method was a survey with closed-ended and open-ended questions, designed to capture both quantitative and qualitative responses to gain a comprehensive understanding of the instructors' views and experiences.

Survey Instrument and its Distribution

The survey instrument was developed based on established survey research practices (Groves et al., 2004), using Google Form, a flexible and widely accessible platform that supports both open-ended and closed-ended question formats. It was designed to elicit detailed early feedback on several key aspects of AI chatbot usage in academic settings from January to March 2024. The survey was disseminated via email to a curated selection of faculty members across various STEM disciplines in higher education.

Survey Questions

The survey included a series of questions designed to capture comprehensive data on instructors' experiences and opinions regarding AI chatbots:

- Demographic Information
- AI Chatbot Usage
- Evaluation of AI Chatbots
- Perceptions and Recommendations

Data Analysis

Numerical data were collected using Likert-scale and multiple-choice questions and analyzed using descriptive statistics in GraphPad Prism version 8.3.0 to identify response patterns and trends. Qualitative data gathered from open-ended questions were analyzed thematically, following the guidelines by Braun and Clarke (2006) to identify common themes, insights, and concerns related to AI integration. This

approach allows for a comprehensive understanding of the views and experiences of instructors.

Sample Description

The instructor sample for this study consists of individuals from diverse academic disciplines and varying levels of teaching experience in Lebanese higher education, particularly within STEM fields. The largest proportion of instructors come from Biological, Chemical, and Food Sciences, representing 46.7% of the sample, followed by Engineering and Computer Science at 20%, and both Public Health (including Biomedical, Nursing, and Nutrition) and Math and Physics at 16.7% each (Table 1). The age distribution reveals that 50% of the instructors are between 30–39 years old, 26.7% are between 40–49 years, 13.3% are between 20–29 years, 6.7% are between 50–59 years, and 3.3% are between 60–69 years. Most instructors hold high academic qualifications, with 70% possessing a Doctorate and 30% holding a Master's degree. In terms of teaching experience, 20% have 1–2 years of experience, 13.3% each have 3–4 years and 5–6 years, 10% have 7–9 years, and another 13.3% each have 10–12 years and 13–15 years of experience, while 16.7% have more than 15 years of experience. This demographic spread underscores a rich mixture of youthful adaptability and seasoned expertise, illustrating the depth and breadth of faculty engagement with AI technologies across various educational and professional stages within the Lebanese STEM academic landscape.

Table 1 Profile characteristics of instructor sample in the study

Category	Percentage
Course Fields	
Biological, Chemical, and Food Sciences	46.7
Engineering and Computer Science	20.0
Public Health (Biomedical, Nursing, Nutrition)	16.7
Math and Physics	16.7
Age Distribution	
30–39 years	50
40–49 years	26.7
20–29 years	13.3
50–59 years	6.7
60–69 years	3.3
Highest Degree Earned	
Doctorate	70
Master's Degree	30
Teaching Experience	
1–2 years	20
3–4 years	13.3
5–6 years	13.3
7–9 years	10.0
10–12 years	13.3
13–15 years	13.3
> 15 years	16.7

Results and Discussion

Faculty Engagement with AI Chatbot Technologies

The adoption of AI chatbots varies significantly across different academic disciplines, reflecting a complex interplay of factors including the nature of the field, the specific demands of the curriculum, and possibly the availability of discipline-specific AI tools. In this study, 'adoption rate' specifically refers to the percentage of instructors who are actively incorporating AI chatbots into their daily activities. A significant portion, 56.7%, of the STEM faculty in this study are already familiar with AI chatbots and have previously used them (Table 2). Despite the close alignment of Engineering and Computer Science with technological innovation, only 18.2% of instructors in these fields use AI chatbots, a notably low figure compared to Biological, Chemical, and Food Sciences. Conversely, 50% do not use AI chatbots, representing the highest non-adoption rate among the categories. This could reflect specific educational needs that current AI technologies do not meet, or perhaps a preference within these fields for developing proprietary tools tailored to their curriculum over adopting generic AI chatbots. The adoption rate among Math and Physics faculty is the lowest among the groups at only 9.1%, with 16.7% not using AI chatbots. The low adoption could be attributed to the abstract and theoretical nature of these fields, where

Table 2 Percentage of AI chatbot usage by instructors across different categories

Category	Yes (56.7)	No (43.3)
Major		
Biological, Chemical, and Food Sciences	45.5	33.3
Engineering and Computer Science	18.2	50.0
Public Health (Biomedical, nursing, etc.)	27.3	0
Math and Physics	9.1	16.7
Age Group		
20–29 years	9.1	0.0
30–39 years	45.5	50.0
40–49 years	45.5	33.3
50–59 years	0	16.7
Teaching Experience		
1–5 years	36.4	16.7
6–10 years	36.4	33.3
11–15 years	18.2	0.0
> 15 years	9.1	50.0

AI applications may not yet be as developed or as useful as in more empirical or data-driven disciplines. This suggests an opportunity for developers to create AI tools specifically designed to address the unique challenges of these disciplines. The Public Health (Biomedical, Nursing, etc.) category shows a 27.3% adoption rate with no instructors reporting non-use of AI chatbots, suggesting that AI chatbots are particularly beneficial and well-integrated within public health education. This promising adoption rate reflects the growing integration of AI in healthcare education, where chatbots are used for clinical simulations, patient case studies, and real-time feedback. Many university instructors across disciplines such as Humanities, Law, Science, and Social Sciences in Sweden acknowledged a limited understanding of AI. However, a significant portion also believed that AI has the potential to foster a more equitable support system for students (McGrath et al., 2023). In another recent study from Germany, most faculty members utilized AI tools in at least two areas, including personal, teaching, or other professional activities. About one-third actively incorporated AI into their teaching, primarily for lesson planning and preparation, while one-quarter had not used AI in their teaching at all (Mah & Groß, 2024).

All these results indicate that while AI chatbots hold significant potential across educational sectors, their adoption is highly contingent on how well they align with the specific needs and workflows of each discipline. The distribution of AI chatbot adoption across different age groups among faculty reveals varied engagement levels, potentially reflecting generational perspectives on technology adoption in academia. The youngest cohort, 20–29 years, shows a modest adoption rate of 9.1%, with no non-users, suggesting either a smaller representation in the sample or minimal resistance to AI chatbot adoption. This finding indicates that younger instructors are generally open to exploring and integrating new technologies into their teaching practices, likely due to greater familiarity and comfort with digital tools (Tsai et al., 2019). In contrast, the 30–39 age group exhibits a substantial engagement at 45.5%, but equally, 50% do not use AI technologies, indicating a polarized view within this demographic, possibly as mid-career educators evaluate AI's benefits against traditional methods. Similarly, 40–49-year-olds report a significant usage rate of 45.5%, with a lower non-adoption rate of 33.3%, suggesting increasing acceptance among more experienced professionals integrating new technologies into established practices. However, the 50–59 age group shows a complete absence of AI adoption, with 16.7% non-users, which could be due to a preference for traditional methodologies, less familiarity with AI technologies, or skepticism about AI's relevance and efficacy. Older adults often face difficulties in learning and adapting to new technologies due to age-related declines in cognitive abilities, which can make interaction with new interfaces more challenging and time-consuming (Schepman & Rodway, 2023; Tsai et al., 2019). This pattern indicates that while younger instructors are exploring AI, there is considerable reluctance among older faculty, highlighting the need for targeted strategies to facilitate AI integration across all age groups to ensure the tools are accessible, relevant, and address specific educational needs effectively. Developing age-specific interventions, such as training workshops tailored to each demographics' unique challenges and perceptions, could foster broader acceptance and meaningful use of AI in academia.

The adoption of AI chatbots across different ranges of teaching experience presents a spectrum of usage patterns, indicating varying levels of acceptance and resistance. Instructors with 1–5 years of teaching experience show a relatively high adoption rate of 36.4%, with only 16.7% not using AI chatbots, suggesting that newer faculty are more inclined to integrate AI into their teaching methods. Similarly, those with 6–10 years of experience also report almost same usage rate of 36.4%, but with a higher non-adoption rate of 33.3%, which may reflect a phase where educators are reassessing the utility of AI tools as they become more established in their careers. Faculty with 11–15 years of experience show an 18.2% adoption rate with no reported non-users, indicating a possible peak in AI acceptance among mid-career educators who likely see the benefits of these technologies. However, those with more than 15 years of experience have the lowest adoption rate at 9.1%, coupled with the highest non-adoption rate of 50%, reflecting significant resistance potentially due to entrenched teaching practices or skepticism about the efficacy and relevance of AI in their longstanding academic routines. This distribution highlights how the integration of AI in educational settings is heavily influenced by the duration of teaching experience, with newer educators more open to adopting AI technologies, while veterans show substantial reluctance, emphasizing the need for strategies that address the specific concerns and support requirements of seasoned faculty. Educators with more years of teaching experience often face challenges such as computer anxiety and a lack of technological skills, which can hinder their ability to integrate new technology effectively. However, senior teachers can still develop a technological-pedagogical-content-knowledge base, which helps them integrate technology despite these barriers (Saienko et al., 2020). Encouraging mentorship programs where experienced educators collaborate with early-career faculty who actively use AI tools could also facilitate knowledge exchange and foster a culture of innovation.

The analysis of AI chatbot usage among academic users reveals that only 17.6% are willing to incur registration fees, highlighting a strong preference for free access to these tools, while 82.4% would not pay, indicating potential barriers in perceived value or budget constraints (Table 3). The

Table 3 AI Chatbot usage and preferences among users

Category	Percentage of users
Registration Fees Willing	
Yes	17.6
No	82.4
Adoption Timing	
1 to 2 months ago	11.8
3 to 6 months ago	17.6
More than 6 months ago	35.3
Haven't used for teaching purposes	35.3
Tried AI-Chatbots	
OpenAI (ChatGPT or DALL-E)	82.4
Bing	11.8
Google Bard (Gemini) & OpenAI	5.9
Most Frequently Used AI-Chatbot	
OpenAI (ChatGPT or DALL-E)	88.2
Bing	11.8
Primary Interaction Platform	
Website	76.5
Phone/Tablet Application	23.5

availability of resources and institutional support are critical factors that positively influence educators' perceptions and adoption of technology. When technology is perceived as easy to use and supported by the institution, educators are more likely to integrate it into their teaching practices (Kurian et al., 2019). To increase adoption among this demographic, providers may need to explore freemium models or emphasize the cost-effectiveness of paid versions through tailored marketing and demonstrations of their impact on academic productivity. When considering the adoption timing, 35.3% of respondents began using AI chatbots for academic purposes over six months ago, showing early adoption, while 29.4% started within the last six months, pointing to ongoing interest. However, 35.3% have not used AI chatbots for teaching purposes at all, which may suggest either a lack of relevance or awareness of these tools in their academic activities. The significant portion of respondents who have not used these tools raises questions about potential gaps in awareness, accessibility, or perceived relevance. Targeted awareness campaigns and training sessions could help bridge these gaps, especially in disciplines where the utility of AI chatbots might not be immediately apparent (Choi et al., 2023).

In terms of usage, OpenAI's ChatGPT or DALL-E dominates with 82.4% of respondents using it frequently, far outpacing Bing and Google Bard (Gemini), which only saw 11.80% and 5.9% usage respectively. This distribution highlights a strong preference for OpenAI's platforms, likely attributed to their advanced features, broad accessibility, and extensive publicity. The primary platforms for interacting with AI chatbots are websites (76.5%) and less for phone/tablet applications (23.5%). This split may reflect several factors: some educators spend a considerable amount of time at computers, phone applications may not be as effective or user-friendly as their website counterparts, or some AI chatbots may lack phone application versions altogether. After completing this study, developers optimized mobile platforms to enhance functionality and boost engagement among users who prefer apps over websites.

The analysis of faculty opinions on AI chatbots, gauged across various academic activities using a Likert scale from 1 (highly unhelpful) to 5 (most helpful), indicates a generally lukewarm reception. This reflects the faculty's existing familiarity and skill levels with AI chatbots. Faculty members find AI chatbots somewhat unhelpful in writing publications and preparing lectures, with mean scores of 2.12 and 2.06 respectively, indicating a gap in AI's ability to support complex scholarly writing and specific pedagogical needs (Fig. 1). The preparation of assessment questions and grading tasks scored 2.18 and 2.00, respectively, suggesting that AI chatbot struggles with generating academically robust questions and consistently grading assessments. Conversely, AI chatbots scored relatively higher in routine tasks like writing emails (2.47) and saving time in daily academic tasks (2.35), showing some efficiency benefits.

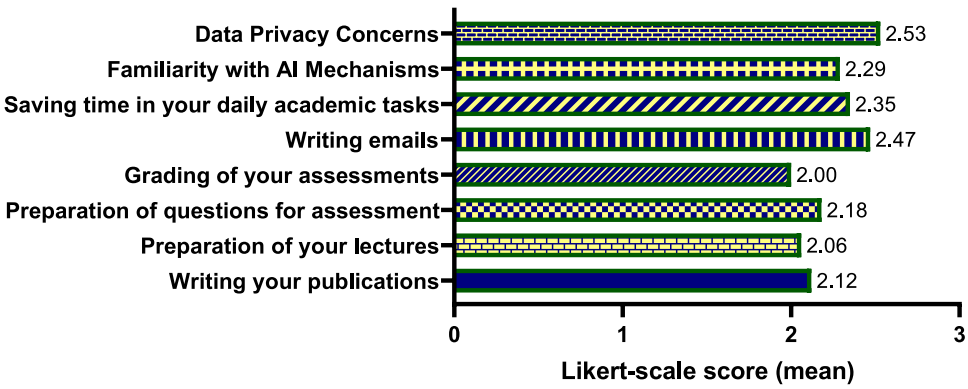


Fig. 1 Likert scale (1 to 5) ratings of AI helpfulness across academic tasks among AI chatbot users

However, these scores still reflect only moderate helpfulness. Familiarity with AI mechanisms scored 2.29, pointing to the potential for improved perceptions with increased understanding of AI functionalities. Data privacy concerns received a higher score of 2.53, underscoring significant worries about the security of sensitive information handled by AI. Given the moderate levels of faculty satisfaction with AI chatbots, particularly in complex academic activities, it is evident that there is a substantial need for targeted training and workshops. These initiatives should aim to enhance faculty proficiency and familiarity with AI technologies, thereby improving their effectiveness in utilizing AI chatbots for both routine and academically demanding tasks. In a recent German study for higher education faculty, AI tools are most commonly used for private purposes and teaching-related activities, with average ratings around 3.19 to 3.75 out of 5. However, the use of AI for direct teaching activities, assessments, and adaptive learning systems is less frequent, with ratings typically below 3 (Mah & Groß, 2024). By addressing the specific gaps in AI support for scholarly writing, pedagogical activities, assessment creation, and grading, such educational programs could significantly bolster the overall utility and acceptance of AI chatbots in higher education settings.

Instructor Encouragement of AI Chatbot Usage Among University Students

In the context of limited regulatory oversight concerning AI in educational institutions at the time of gathering data, the survey results regarding instructors' attitudes towards AI chatbot usage among students at various academic levels provide a detailed insight into the integration of AI in education. Each academic tier—Freshman, Sophomore, Junior, and Graduate—exhibits distinct response patterns, underscoring the diverse levels of acceptance and the array of concerns related to AI technology (Fig. 2). For freshman undergraduates, instructors show a diverse range of opinions with 17.6% discouraging AI use and an equal percentage strongly discouraging it, reflecting significant reservations about introducing AI to students early in their academic journey. However, 23.5% encourage its use and 5.9% strongly encourage it, suggesting that some instructors see value in early exposure to AI tools. A notable 35.3% remain neutral, indicating uncertainty or conditional acceptance of AI's role for freshmen. For sophomore undergraduates, the responses among instructors show a higher discouragement rate at 29.4%, and a complete absence of strong encouragement. Yet, the same percentage (29.4%) of instructors encourage AI use, and another 29.4% remain neutral. This suggests a critical point in the curriculum where instructors are possibly evaluating the benefits of AI against its potential drawbacks. For junior undergraduates, instructors display a notably high level of encouragement, with 52.9% endorsing the use of AI. Only a small fraction, 5.9%, strongly discourage its use, and 11.8% express general discouragement. This significant level of encouragement suggests that as students acquire foundational knowledge and mature academically, instructors grow increasingly confident in the benefits of AI chatbots, viewing these students as well-prepared to effectively utilize AI tools. At the graduate level, the encouragement of AI chatbots use remains high at 41.2%, with a return to stronger encouragement (5.9%) not seen since the freshman level. This suggests that for advanced studies, the potential of AI to assist with complex, specialized tasks is recognized. However, the rates of discouragement and strong discouragement (5.9% and 11.8%, respectively) alongside a significant neutral stance (35.3%) reflect ongoing

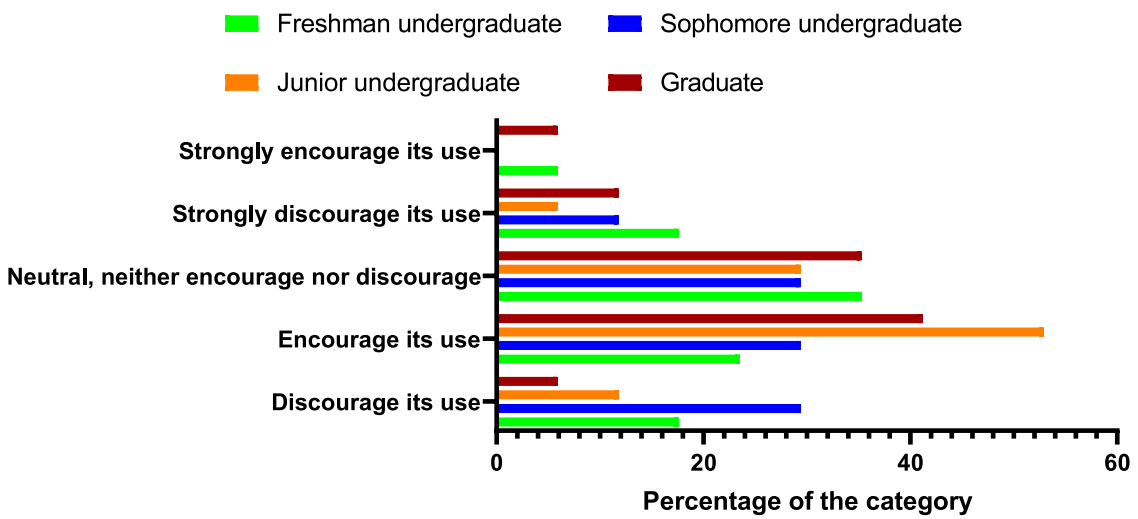


Fig. 2 Instructor Responses on Student AI Chatbot Encouragement Based on Academic Level

concerns about the depth and reliability of AI applications in high-stakes academic work. The persistence of neutral and discouraging views at all levels underscores lingering doubts about AI's reliability, its potential to foster over-reliance, and ethical considerations, which highlights the complex balance instructors are attempting to strike between leveraging innovative tools and maintaining rigorous academic standards. On the other hand, a study involving Portuguese graduate students found that more than half of the participants reported using ChatGPT, primarily for academic purposes such as searching for information and generating initial ideas for topics (Costa et al., 2024). Another survey of 473 undergraduate students at King Faisal University revealed that 60% used ChatGPT for homework assistance, 45.5% for research, and 35.5% for exam preparation (Almulla & Ali, 2024).

In assessing faculty opinions on whether AI chatbots will enhance human intelligence, the survey responses reflect a generally cautious and yet complex view. A notable 23.53% of respondents agree on the potential of AI chatbots to enhance human intelligence, while the largest segment at 35.29% disagrees with this potential, indicating deep reservation (Fig. 3). A significant 23.53% of respondents remain neutral, reflecting uncertainty possibly due to the current limitations of AI technologies or a lack of conclusive evidence. Additionally, 17.65% strongly disagree, indicating skepticism about AI's ability to understand and augment human cognitive processes or concerns about overreliance on technology. Interestingly, no respondents strongly agree, suggesting a need for more demonstration of AI's effectiveness in cognitive enhancement. Overall, the mixed responses underscore a cautious yet open perspective among faculty toward the integration of AI chatbots in enhancing human intelligence, highlighting both the potential benefits and significant challenges that must be addressed for wider acceptance and effective use.

Faculty Perspectives on the Pros and Cons of AI Chatbots in University Settings

The summary of instructor feedback on the advantages and disadvantages of AI chatbot for university students illustrates a nuanced perspective on their integration in academic settings (Table 4). Advantages highlighted include 24/7 accessibility, which enhances learning outside classroom hours, efficiency in research and information retrieval, and valuable support in synthesizing complex materials and aiding complex problem-solving in projects. However, these benefits are tempered by significant concerns: there's a risk of students becoming overly dependent on technology, potentially

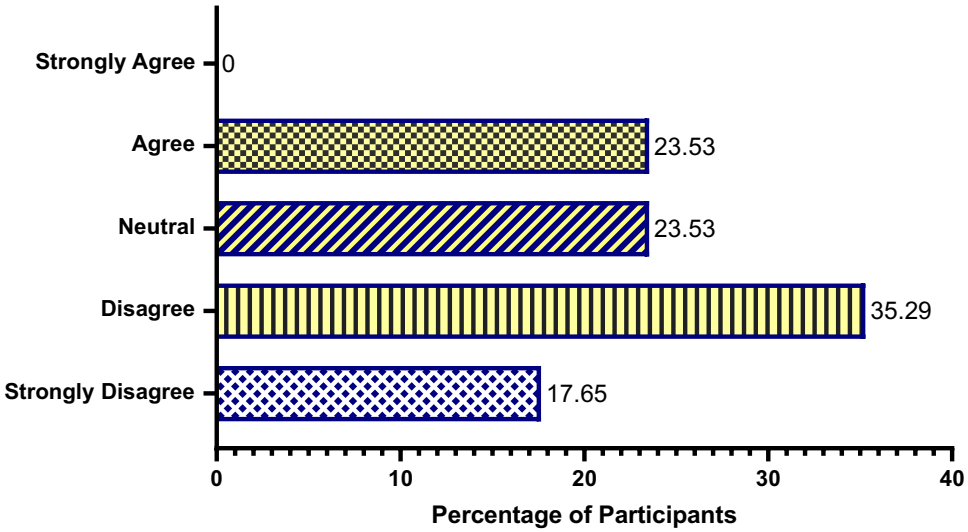


Fig. 3 Faculty opinions on the potential of AI chatbots to enhance human intelligence expressed as percentages

Table 4 Overview of advantages and disadvantages of AI chatbot for university students based on instructor feedback

Category	Advantages	Disadvantages
Accessibility	Available 24/7, providing constant assistance	Dependency on technology for basic tasks
Efficiency	Speeds up research and access to information	May overlook nuanced or context-specific info
Research Quality	Helps synthesize complex data and texts	Potential for spreading misinformation
Project Support	Aids in solving complex problems and tasks	Could reduce critical thinking skills
Skepticism	-	Concerns about data privacy and accuracy

diminishing their ability to handle academic tasks independently and leading to a reduction in critical thinking skills. Additionally, AI chatbot might overlook nuanced information and spread misinformation, with further worries about data privacy and the accuracy of information. Based on the literature, AI chatbots like ChatGPT provide significant benefits for educators by automating tasks such as scheduling and grading, thereby freeing up time for instructional planning and student interaction. Moreover, these chatbots enhance teaching by producing personalized learning materials and activities tailored to individual student needs. While AI cannot replace human instructors entirely, its integration promises to ease educators' workload and enhance instructional quality. Conversely, scholars have expressed concerns about the integration of AI chatbots in education, particularly regarding reliability and accuracy, fair assessments, and ethical implications (Labadze et al., 2023).These mixed views underscore the need for a balanced approach in incorporating AI chatbot into educational frameworks, emphasizing robust AI governance and ongoing evaluation to ensure these tools support educational goals without compromising critical academic skills.

Conclusion

The investigation into the early perceptions of AI chatbots' role in higher education within Lebanon's STEM faculties has effectively addressed the research questions posed at the outset of this study. Firstly, our findings detail how rapidly faculty members are adapting to and engaging with recent AI technologies. The adoption rates vary significantly by age, major, and years of teaching experience, revealing a trend where younger and less experienced faculty are quicker to integrate AI into their practices. Secondly, we have identified specific benefits that Lebanese STEM faculty have recognized from using AI technologies. These include continuous student support, streamlined research processes, and enhanced assistance with academic projects, which have been praised for their ability to improve educational outcomes and operational efficiency. However, the study also uncovered principal challenges and opportunities, with educators expressing concerns about data privacy, the integrity of academic work, and the potential erosion of critical thinking skills due to over-reliance on AI. These concerns have led to calls for the development of training programs aimed at enhancing faculty understanding and proficiency with AI technologies. Importantly, this study has highlighted that the enthusiastic reception of AI is moderated by significant caution and skepticism, often stemming from a lack of familiarity with these tools. While the conclusions are based on data gathered from 30 STEM faculty members, their limited generalizability, due to the small sample size, nonetheless provides deep insights into the complex dynamics of integrating AI technologies in higher education. Furthermore, the study's regional focus on Lebanon may not accurately represent global trends in STEM faculties. The reliance on self-reported perceptions also introduces potential biases, as participants' responses might not fully reflect their actual practices. These limitations should be carefully considered when interpreting the findings. Future research employing a mixed-methods approach could extend these findings, offering a more comprehensive and generalized understanding of AI's impact on educational practices in Lebanon.

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Data Availability The data from this study can be obtained by contacting the corresponding author directly. Due to ethical considerations and to maintain the confidentiality agreed upon with the participants, the data is not publicly accessible.

Declarations

Conflicts of Interest The author declares no conflict of interest.

Declaration of Generative AI and AI-assisted Technologies in the Writing Process The authors used ChatGPT to review and refine the language in this document. After using the tool, they thoroughly revised the content and took full responsibility for the publication.

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